Smart Campus Management System

FINAL YEAR COMPUTER SCIENCE PROJECT

A web-based platform transforming how universities manage operations, communication, and academic workflows.

2025–2026 | Computer Science

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What is SCMS?

The Platform

A web-based smart ecosystem that automates and centralizes all campus management operations — replacing paper-based, manual workflows with a unified digital experience.

Who Uses It?

🛡️ Admins

Full control

👩‍🏫 Faculty

Course ops

🎓 Students

Daily access

Real-World Problem Solved

Universities suffer from fragmented systems, attendance inefficiencies, and slow approvals. SCMS unifies it all into one secure platform.

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Problem Statement

Current campus management practices are fragmented, manual, and error-prone — creating bottlenecks across every department.

Manual Attendance Tracking

Time-consuming roll calls and paper registers introduce errors and require hours of reconciliation.

No Centralized Platform

Students, faculty, and administration operate in silos with no unified system connecting them.

Inefficient Communication

Announcements and updates are scattered across emails, notice boards, and informal channels.

No Role-Based Access Control

Sensitive data is accessible without proper security boundaries between user roles.

Paper-Based Document Flows

Assignments and reports still handled via email and physical documents — slow and untrackable.

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Project Objectives

SCMS is designed to solve each of these challenges through a set of clearly defined, measurable goals.

Automate Operations

End-to-end automation of campus workflows

Single Platform

One unified digital space for all roles

Role-Based Security

Admin / Faculty / Student access control

Better Communication

Targeted, real-time announcements for all roles

Digital Attendance

QR-code based session attendance tracking

Assignment Management

Upload, submit, and manage reports digitally

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System Overview

WEB-BASED APPLICATION — REACT + NODE.JS + MONGODB

Three distinct roles interact with a unified backend through a secure REST API architecture.

🛡️ **Admin**	Manage faculty and students, post announcements, approve/reject registrations, view all system records
👩‍🏫 **Faculty**	Create QR attendance sessions, manage and grade assignments, post role-specific announcements, approve student enrollment
🎓 **Student**	Register with campus code, scan QR codes for attendance, submit assignments, view schedules and announcements

All three roles connect through a secure, JWT-authenticated, API-driven backend built on the MERN stack.

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Key Features

Admin Dashboard

Real-time statistics, user lists, and full system control in a single view.

Faculty Management

Add, validate, and manage faculty profiles with approval workflows.

Student Management

Enrollment, profile management, and tiered approval for student access.

QR-Based Attendance

Time-limited, session-specific QR codes for live attendance marking.

Assignment & Report Management

Upload, view, and download assignments with faculty review workflows.

Role-Based Announcements

Targeted notifications delivered to specific user groups only.

Approval Workflows

3-layer verification chain: Admin ↔️ Faculty ↔️ Student.

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What Makes SCMS Different

Beyond standard features, SCMS is built with mechanisms that ensure security, integrity, and institutional trust.

Campus Code Registration

Only users with a valid campus code can register — preventing unauthorized access at the entry point.

Role-Based JWT Authentication

JSON Web Tokens enforce strict role separation at every API layer, ensuring data never crosses role boundaries.

3-Layer Approval Chain

Admin validates faculty → Faculty validates students → Full access granted only after all verifications pass.

Time-Limited QR Sessions

QR codes expire after the session window — preventing proxy attendance and ensuring authenticity.

Scalable REST Architecture

Built with data validation at every layer — designed to scale as the institution grows.

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Technology Stack

Built on the MERN Stack

Each layer was chosen for performance, scalability, and developer ecosystem maturity. All components communicate via RESTful API conventions.

🎨 **Frontend**	React.js — component-based UI
⚙️ **Backend**	Node.js + Express.js — REST API server
🗄️ **Database**	MongoDB (NoSQL) — flexible data modeling
🔐 **Auth**	JWT — stateless, role-secured tokens
🛠️ **Dev Tools**	Git, Postman, VS Code

Why MERN?

- JavaScript across the entire stack — unified language
- MongoDB's document model maps naturally to campus data
- React enables fast, reactive dashboard UIs
- Express.js provides lightweight, high-performance routing
- Massive open-source ecosystem and community support

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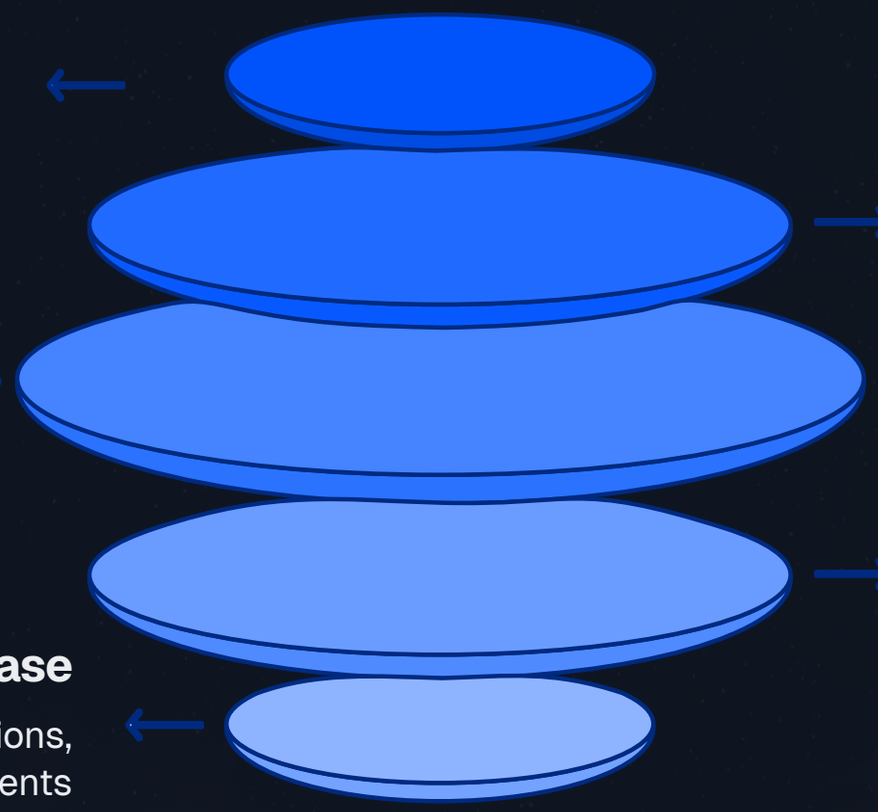
System Architecture

A layered, API-driven architecture ensures clean separation of concerns, secure data flow, and scalable design across all three user roles.



React Frontend

Client UI with role-based views

REST API Layer

HTTP/HTTPS endpoints returning JSON

Node.js Backend

Business logic with JWT checks

Mongoose ODM

Schema validation and data models

MongoDB Database

Stores users, sessions, assignments

JWT tokens are verified at every API request — the backend never trusts the client without authentication validation.

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Core Modules

🛡️ Admin Module

- Manage all users system-wide
- Approve or reject faculty & student registrations
- Post institution-wide announcements
- View all records and system activity

👩‍🏫 Faculty Module

- Create and manage QR attendance sessions
- Assign, grade, and review student work
- Post role-targeted announcements
- Approve student enrollment requests

🎓 Student Module

- Register using unique campus code
- Scan QR codes to mark live attendance
- Submit assignments and view feedback
- Access schedules and announcements

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